

Scanner Scoring Engine V2: Discovery Mode — Final Production Spec

Objective

Maximize the probability that a top-ranked ticker experiences a close-to-close move of at least 3% within the next 5-10 trading days, conditional on passing the liquidity gate. Both bullish and bearish pre-move setups are scored equally. The scanner is direction-agnostic in scoring but tags each result with a directional lean for downstream use by the Strategist.

All bucket thresholds in this spec are V1 priors and will be empirically re-optimized after the backtest described in the Calibration section. All thresholds are stored in config, not hardcoded. No code changes required to adjust them.

Architecture

Total score: 100 points across 5 categories. Options Liquidity is a pre-filter gate applied before scoring.

Weights (initial priors, empirically calibrated after backtesting):

- Setup Quality: 20
- Accumulation Pattern: 15
- IV Setup: 25
- Flow Divergence: 25
- Emerging RS: 15

IV Setup and Flow Divergence are weighted highest because they are the most forward-looking signals.

Pre-Filter: Options Liquidity Gate

Applied before any scoring. Tickers that fail are excluded entirely.

Calculation:

- Pull near-ATM options (strikes within 10% of spot) for expirations 20-60 DTE

- $\text{avg_spread_pct} = \text{average}((\text{ask} - \text{bid}) / \text{mid})$ across those strikes
- total_oi = sum of open interest across those strikes

Scoring (0-15):

- $\text{avg_spread_pct} \leq 2\%$: 15
- $\text{avg_spread_pct} > 2\%$ and $\leq 5\%$: 12
- $\text{avg_spread_pct} > 5\%$ and $\leq 10\%$: 9
- $\text{avg_spread_pct} > 10\%$ and $\leq 15\%$: 6
- $\text{avg_spread_pct} > 15\%$: 0

Hard gate: score must be ≥ 8 to be included. Below 8 = excluded. Scores 8-15 serve as tiebreaker only when composite scores are equal. Liquidity score is NOT added to the 100-point composite.

Pre-Filter: Guardrails

Exclude before scoring:

- Stock price below \$10
- 20-day average daily volume below 500,000 shares
- Market cap below \$2 billion
- Primary-listing ADRs with 20-day average volume below 1,000,000
- IPOs with fewer than 60 trading days of history (insufficient data for SMA, ATR, OBV, RS)
- Stocks halted during the scan window
- Stocks that split within the last 10 trading days (distorted volume and price data)

These thresholds are configurable in the system config.

Category 1: SETUP QUALITY (20 points)

Measures whether the stock is positioned for a move, not already in one.

Data sources: Schwab equity data (price, SMA, ATR), Market Pulse bias.

Sub-score 1A: Price Distance from 20-Day SMA (0-7 points)

Formula: $d = \text{abs}(\text{price} - \text{SMA}_{20}) / \text{SMA}_{20} * 100$

Buckets (strict inequalities on lower bounds):

- $d \leq 1.0\%$: 7 points
- $d > 1.0\%$ and $\leq 3.0\%$: 5 points
- $d > 3.0\%$ and $\leq 6.0\%$: 3 points
- $d > 6.0\%$ and $\leq 10.0\%$: 1 point
- $d > 10.0\%$: 0 points

Sub-score 1B: Range Compression via ATR Ratio (0-7 points)

Formula: $RC = \text{ATR}_5 / \text{ATR}_{20}$

Buckets:

- $RC \leq 0.50$: 7 points
- $RC > 0.50$ and ≤ 0.65 : 5 points
- $RC > 0.65$ and ≤ 0.80 : 3 points
- $RC > 0.80$ and ≤ 1.00 : 1 point
- $RC > 1.00$: 0 points

Sub-score 1C: Pulse Directional Alignment (0-6 points)

Pulse enum values used here match the existing v1 Pulse output (BULLISH, MODERATELY_BULLISH, NEUTRAL, MODERATELY_BEARISH, BEARISH). No changes to Pulse output format.

Logic:

- Pulse is BULLISH or MODERATELY_BULLISH and $\text{price} > \text{SMA}_{20}$: 6 points
- Pulse is BEARISH or MODERATELY_BEARISH and $\text{price} < \text{SMA}_{20}$: 6 points
- Pulse is NEUTRAL: 3 points
- Pulse direction conflicts with price position: 2 points (small penalty, not zero, because misaligned setups can still be valid micro-override candidates)

Category 2: ACCUMULATION PATTERN (15 points)

Measures whether volume is building quietly without proportional price movement.

Data sources: Schwab equity data (price, volume). OBV is computed (see below).

Sub-score 2A: Volume Trend Ratio (0-6 points)

Formula: $VR = \text{median}(\text{volume}, 5d) / \text{median}(\text{volume}, 20d)$

Use median to reduce sensitivity to blowout days.

Buckets:

- $VR \geq 1.2$ and ≤ 2.0 : 6 points
- $VR > 1.0$ and < 1.2 : 3 points
- $VR > 2.0$ and ≤ 3.0 : 2 points
- $VR \leq 1.0$: 0 points
- $VR > 3.0$: 0 points

Sub-score 2B: Volume-Price Divergence (0-5 points)

Conditional logic, not a division formula.

Inputs:

- $5d_price_change_pct = (\text{close_today} - \text{close_5d_ago}) / \text{close_5d_ago} * 100$
- VR from sub-score 2A

Scoring (strict inequalities):

- IF $VR \geq 1.2$ AND $5d_price_change_pct > -2.0\%$ and $\leq +2.0\%$: 5 points
- IF $VR \geq 1.2$ AND $5d_price_change_pct > +2.0\%$ and $\leq +4.0\%$: 3 points
- IF $VR \geq 1.2$ AND $5d_price_change_pct \geq -4.0\%$ and $< -2.0\%$: 3 points
- IF $VR \geq 1.2$ AND $\text{abs}(5d_price_change_pct) > 4.0\%$: 0 points
- IF $VR < 1.2$: 0 points

Sub-score 2C: OBV Direction (0-4 points)

OBV must be computed as a daily running total: if today's close > yesterday's close, add today's volume. If close < yesterday's, subtract. If equal, add 0.

OBV_slope = linear regression slope of OBV over last 10 trading days, normalized by

dividing by 20-day average daily volume.

Scoring:

- OBV_slope positive AND 10d_price_change_pct between -2% and +2%: 4 points (accumulation)
- OBV_slope negative AND 10d_price_change_pct between -2% and +2%: 3 points (distribution, also tradeable)
- OBV_slope direction matches price direction: 1 point (confirmation)
- OBV flat: 0 points

"Flat" defined as $\text{abs}(\text{normalized_OBV_slope}) < 0.05$.

We slightly favor accumulation (4 pts) over distribution (3 pts) because the majority of actionable setups in this system are long-biased due to defined-risk vertical spreads.

Category 3: IV SETUP (25 points)

The highest-conviction leading signal category.

Data sources: Schwab options chain. Anchor tenor: 30-day implied volatility from the nearest monthly expiration with 20-40 DTE.

HV_20d definition: If Schwab provides HV20, use it. Otherwise compute as annualized 20-day historical volatility using the standard formula: $\text{stdev}(\ln(\text{close}_t / \text{close}_{t-1}))$ over 20 trading days, multiplied by $\text{sqrt}(252)$.

Sub-score 3A: IVR Level (0-7 points)

Buckets:

- IVR ≥ 70 : 7 points
- IVR ≥ 50 and < 70 : 5 points
- IVR ≥ 35 and < 50 : 3 points
- IVR ≥ 20 and < 35 : 1 point
- IVR < 20 : 0 points

Sub-score 3B: IVR 5-Day Change (0-10 points)

Formula: $\text{IVR_change} = \text{IVR_today} - \text{IVR_5d_ago}$

Buckets:

- IVR_change ≥ 15 : 10 points
- IVR_change ≥ 10 and < 15 : 8 points
- IVR_change ≥ 7 and < 10 : 6 points
- IVR_change ≥ 3 and < 7 : 4 points
- IVR_change ≥ 1 and < 3 : 2 points
- IVR_change < 1 : 0 points

EARNINGS SUPPRESSION: If the ticker has earnings within 14 calendar days, cap this sub-score at 4 points maximum. Rising IVR near earnings is scheduled, not a discovery signal.

Lookback is parameterized in config (default 5 days, tunable to 3 or 10 after backtesting).

Sub-score 3C: IV vs HV Gap (0-8 points)

Formula: $iv_hv_ratio = IV_{30d} / HV_{20d}$

Buckets:

- $iv_hv_ratio \geq 1.50$: 8 points
- $iv_hv_ratio \geq 1.30$ and < 1.50 : 6 points
- $iv_hv_ratio \geq 1.15$ and < 1.30 : 4 points
- $iv_hv_ratio \geq 1.05$ and < 1.15 : 2 points
- $iv_hv_ratio < 1.05$: 0 points

EARNINGS SUPPRESSION: If earnings within 14 days, cap at 3 points maximum.

Category 4: FLOW DIVERGENCE (25 points)

Data source: Polygon.io options flow.

MISSING DATA RULE: If Polygon data is missing, unavailable, or stale (>24 hours since last full trading day's data) for a ticker, this entire category scores as N/A. Do NOT score 0.

Exclude Flow Divergence from the composite and renormalize: $composite = (other\ categories\ total) / 75 * 100$. Display "Flow data unavailable - renormalized score" badge on the scanner result card.

"Today" for flow data means the last completed trading day. Scans reference end-of-day flow data.

Strike/expiry filter: only consider strikes within 20% of spot and expirations within 60 DTE.

Sub-score 4A: Volume/OI Ratio (0-8 points)

For each qualifying strike with OI ≥ 200 , compute $\text{vol_oi} = \text{daily_volume} / \text{open_interest}$. Take the notional-weighted average across all qualifying strikes.

Minimum OI threshold: 200 contracts per strike. Skip strikes below this.

Buckets:

- $\text{avg_vol_oi} \geq 0.80$: 8 points
- $\text{avg_vol_oi} \geq 0.50$ and < 0.80 : 6 points
- $\text{avg_vol_oi} \geq 0.30$ and < 0.50 : 4 points
- $\text{avg_vol_oi} \geq 0.15$ and < 0.30 : 2 points
- $\text{avg_vol_oi} < 0.15$: 0 points

Sub-score 4B: Put/Call Volume Skew (0-5 points)

Formula: $\text{pc_skew} = \text{abs}(\text{call_volume} - \text{put_volume}) / (\text{call_volume} + \text{put_volume})$

ZERO VOLUME GUARD: If $(\text{call_volume} + \text{put_volume}) == 0$, score = 0. Do not attempt the division.

Scores the magnitude of skew, not direction. Both bullish and bearish conviction are tradeable.

Buckets:

- $\text{pc_skew} \geq 0.70$: 5 points
- $\text{pc_skew} \geq 0.50$ and < 0.70 : 4 points
- $\text{pc_skew} \geq 0.30$ and < 0.50 : 2 points
- $\text{pc_skew} < 0.30$: 0 points

Sub-score 4C: Block Trade Detection (0-5 points)

A block trade meets BOTH conditions:

- Notional value $> 0.5\%$ of the ticker's 20-day average daily options notional
- Contract count > 500

Definition of "20-day average daily options notional": $\text{sum of } (\text{daily_volume} * \text{mid_price} * \text{notional_value})$

100) across all strikes and expirations, averaged over 20 trading days. Use Polygon aggregate data.

Buckets:

- 3+ qualifying blocks today: 5 points
- 2 qualifying blocks: 3 points
- 1 qualifying block: 2 points
- 0 blocks: 0 points

Sub-score 4D: Flow vs Price Divergence (0-7 points)

Determine flow_direction:

- $\text{call_volume} > 1.5 * \text{put_volume}$: BULLISH
- $\text{put_volume} > 1.5 * \text{call_volume}$: BEARISH
- otherwise: NEUTRAL

ZERO VOLUME GUARD: If $\text{call_volume} == 0$ and $\text{put_volume} == 0$, flow_direction = NEUTRAL.

Determine price_direction from 5d_price_change_pct:

- $\geq +2\%$: UP
- $< -2\%$: DOWN
- between -2% and $+2\%$ (inclusive): FLAT

Scoring:

- Flow BULLISH and price FLAT or DOWN: 7 points
- Flow BEARISH and price FLAT or UP: 7 points
- Flow BULLISH and price UP: 2 points
- Flow BEARISH and price DOWN: 2 points
- Flow NEUTRAL: 0 points

Category 5: EMERGING RELATIVE STRENGTH (15 points)

Data sources: Schwab equity data for ticker and SPY. Sector ETF mapping table for sector divergence.

Sub-score 5A: RS Momentum (0-8 points)

Formula:

- $RS_ratio = ticker_price / SPY_price$ (daily series, 60-day rolling for context)
- RS_5d_slope = linear regression slope of RS_ratio over last 5 trading days
- RS_20d_slope = linear regression slope of RS_ratio over last 20 trading days
- Normalize slopes by dividing by average RS_ratio value over respective period

"Flat" defined as $abs(normalized_slope) < 0.001$.

Scoring (symmetric for bullish and bearish turns):

- RS_5d_slope positive AND RS_20d_slope negative or flat: 8 points (bullish turn from weakness)
- RS_5d_slope negative AND RS_20d_slope positive or flat: 8 points (bearish turn from strength)
- RS_5d_slope positive AND RS_20d_slope positive: 3 points (established bullish RS)
- RS_5d_slope negative AND RS_20d_slope negative: 3 points (established bearish RS)
- RS_5d_slope flat: 0 points

Sub-score 5B: Sector Divergence (0-7 points)

Sector mapping: each ticker mapped to its GICS sector ETF (XLK, XLF, XLE, XLV, XLI, XLC, XLY, XLP, XLU, XLRE, XLB). This is a static lookup table that needs to be built and maintained. Refresh cadence: quarterly or when holdings change.

FALLBACK: If sector ETF mapping is unavailable for a ticker, use SPY as the benchmark.

Formula:

- $ticker_5d_change$ = 5-day price change %
- $sector_5d_change$ = 5-day price change % of sector ETF
- $divergence = ticker_5d_change - sector_5d_change$

Scoring (symmetric, both directions are tradeable):

- $abs(divergence) \geq 3.0\%$: 7 points
- $abs(divergence) \geq 1.5\%$ and $< 3.0\%$: 4 points
- $abs(divergence) \geq 0.5\%$ and $< 1.5\%$: 2 points

- $\text{abs}(\text{divergence}) < 0.5\%$: 0 points

Directional Lean Tagging

After scoring, compute a directional lean for each result. This is NOT part of the score. It is a tag displayed on the scanner result card and passed to the Strategist for strategy selection.

Direction-sensitive sub-scores and their signals:

1. Pulse alignment (1C): Pulse bias direction
2. OBV slope (2C): positive = bullish, negative = bearish
3. Flow skew direction (4B): raw call_volume > put_volume = bullish, inverse = bearish
4. Flow vs price direction (4D): flow_direction value
5. RS 5d slope direction (5A): positive = bullish, negative = bearish

Count bullish signals and bearish signals from these 5 inputs.

If Flow Divergence is N/A for a ticker (missing Polygon data), treat signals 3 and 4 (flow skew and flow vs price) as NEUTRAL when counting bullish/bearish. The directional lean is then computed from the remaining 3 signals: 3+ in one direction = that tag, otherwise MIXED.

Tagging:

- 4+ bullish signals: tag BULLISH
- 4+ bearish signals: tag BEARISH
- 3 or fewer in either direction: tag MIXED

Display the tag on the scanner card next to the score (e.g., "INTC 85 BULLISH" or "XOM 78 MIXED").

The Strategist uses this tag: BULLISH tag biases toward bull call spreads / bull put spreads. BEARISH tag biases toward bear put spreads / bear call spreads. MIXED tag requires the Strategist to evaluate further before selecting direction.

IMPORTANT: Conflicting signals do NOT reduce the composite score. A stock with strong bullish flow AND strong bearish RS gets full points for both. The tension itself is the setup. The directional tag helps the Strategist interpret it, but the scanner's job is to find tension, not resolve it.

Composite Score Calculation

$\text{composite} = \text{setup_quality} + \text{accumulation} + \text{iv_setup} + \text{flow_divergence} + \text{emerging_rs}$

If flow_divergence is N/A (missing Polygon data): $\text{composite} = (\text{setup_quality} + \text{accumulation} + \text{iv_setup} + \text{emerging_rs}) / 75 * 100$

Round to nearest integer. Display as score out of 100.

Micro-override logic remains unchanged from v1 and will now trigger on early Discovery setups rather than extended momentum names.

Minimum Score Threshold

Initial threshold: 55 for V1. Recalibrate after observing score distribution under Discovery mode. Threshold is stored in config.

Migration

- Discovery mode becomes the default scoring preset
- Current v1 scoring preserved as "Momentum" preset accessible via Scanner settings
- Scanner UI shows active mode: "DISCOVERY" or "MOMENTUM" label
- Deploy Discovery with a visible "BETA" tag until historical backtest validates performance. Remove BETA after validation.

Calibration and Validation Plan

Phase 1: Historical backtest (before removing BETA tag)

- Run Discovery and Momentum scoring in parallel against at least 12 months of historical daily data across the full scan universe
- For each day's top 10 results from each scoring mode, measure:
 - % that moved $\geq 3\%$ (either direction) within 5 trading days
 - % that moved $\geq 3\%$ within 10 trading days
 - Average forward 5d and 10d return
 - Average forward 10d return per unit of risk (Sharpe-style)
 - % that went nowhere ($< 1\%$ move in 10 days) = false positive rate
 - Sector concentration of top 10 results

- Discovery must outperform Momentum on hit rate and false positive rate to be promoted from BETA to default

Phase 2: Live shadow (concurrent with Phase 1 if backtest infra not ready)

- Run both scoring modes live for minimum 30 trading days
- Track same metrics as Phase 1 on live data
- If backtest infrastructure is not feasible before launch, deploy Discovery as BETA with Momentum toggle available and require Phase 1 backtest before removing BETA

Data Dependencies

Indicator	Source	Available	New Build
Price, SMA, ATR	Schwab equity stream	Yes	No
Volume (daily)	Schwab equity stream	Yes	No
OBV	Computed from price + volume	No	Yes (daily running total)
IVR, IV_30d	Schwab options chain	Yes	No
HV_20d	Compute from Schwab daily prices: $\text{stdev}(\ln(\text{close}_t/\text{close}_{t-1}), 20d) * \text{sqrt}(252)$	Verify	Build if missing
Options flow	Polygon.io	Yes	No
SPY price data	Schwab equity stream	Yes	No
Sector ETF mapping	New static lookup table	No	Yes (11 GICS sector ETFs)
Sector ETF prices	Schwab equity stream	Need to subscribe	Add symbols to feed
Earnings dates	Already in system	Yes	No

Edge Cases

- Earnings within 14 days: IV sub-scores 3B and 3C capped per earnings suppression rules. All other categories score normally.
- Stock halted: exclude entirely.
- ETFs in universe: score normally.
- IPOs < 60 trading days: exclude.
- Stock split < 10 trading days: exclude.
- HV_20d = 0 (theoretically possible on completely flat stock): set iv_hv_ratio to 0, sub-score 3C = 0. Do not divide by zero.
- All qualifying strikes have OI < 200 in Flow category: treat Flow as N/A, renormalize.

Indicator Reference Table

Sub-Score	Category	Max Pts	Formula	Lookback
1A	Setup Quality	7	$\text{abs}(\text{price} - \text{SMA20}) / \text{SMA20}$	20d
1B	Setup Quality	7	$\text{ATR5} / \text{ATR20}$	5d, 20d
1C	Setup Quality	6	Pulse bias vs price position	Current
2A	Accumulation	6	$\text{median}(\text{vol}, 5\text{d}) / \text{median}(\text{vol}, 20\text{d})$	5d, 20d
2B	Accumulation	5	VR + price change conditional	5d
2C	Accumulation	4	OBV 10d slope normalized	10d
3A	IV Setup	7	IVR level	52-week
3B	IV Setup	10	IVR today - IVR 5d ago	5d
3C	IV Setup	8	$\text{IV30d} / \text{HV20d}$	30d, 20d
4A	Flow Divergence	8	notional-weighted avg vol/OI	Daily
4B	Flow Divergence	5	$\text{abs}(\text{call-put}) / \text{total}$	Daily

4C	Flow Divergence	5	Block count by relative threshold	Daily
4D	Flow Divergence	7	Flow direction vs price direction	5d, daily
5A	Emerging RS	8	RS ratio 5d slope vs 20d slope	5d, 20d
5B	Emerging RS	7	Ticker 5d - sector ETF 5d	5d

What This Spec Does NOT Cover

- Backtesting infrastructure implementation
- Real-time intraday re-scoring
- AI narrative changes for scanner results
- UI changes beyond Discovery/Momentum toggle, directional lean tag, and flow data badge
- Short-squeeze filtering (if a squeeze setup scores well on these metrics, it surfaces naturally)